

EXDA Metadata Report

Project: VH2D-Project

Generated: 2026-05-24 18:08:56

Includes: Plan, DAQ Systems, Sensors Mapping, Gas Mixing

Responsible Researcher: PhD Student Javier I. Camacho



EXDA Plan Export - Experiment_Plan_v000

Responsible Researcher: PhD Student Javier I. Camacho

Objective: LU / DBI / Kistler pressure DAQ systems characterisation.

Start: 2026-05-25 | Deadline: 2026-05-29

Total Experiments: 18

Generated: 2026-05-24 18:08:55

Group: Prep

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
Prep-01	No	2026-05-25	12	back	Thin Plas.	101325	293	6	Prep-01/Prep-01-DAQ-1 Prep-01/Prep-01-DAQ-2 Prep-01/Prep-01-DAQ-3 Prep-01/Prep-01-H2CM-1 Prep-01/Prep-01-H2CM-2 Prep-01/Prep-01-H2CM-3
Prep-02	No	2026-05-25	12			101325	293	0	-

Group: VH2D-01

Group Objective: Vented deflagration concentration series (Hotwire)

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-01-01	No	2026-05-26	12	Back	Thin Plas.	101325	293	0	-
VH2D-01-02	No	2026-05-26	12			101325	293	0	-
VH2D-01-03	No	2026-05-26	14			101325	293	0	-
VH2D-01-04	No	2026-05-26	14			101325	293	0	-
VH2D-01-05	No	2026-05-26	16			101325	293	0	-
VH2D-01-06	No	2026-05-26	16			101325	293	0	-
VH2D-01-07	No	2026-05-26	18			101325	293	0	-
VH2D-01-08	No	2026-05-26	18			101325	293	0	-

Group: VH2D-02

Group Objective: Vented deflagration concentration series (NO Hotwire)

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-02-01	No	2026-05-26	12			101325	293	0	-
VH2D-02-02	No	2026-05-26	14			101325	293	0	-
VH2D-02-03	No	2026-05-26	16			101325	293	0	-

VH2D-02-04	No	2026-05-26	18			101325	293	0	-
------------	----	------------	----	--	--	--------	-----	---	---

Group: VH2D-03

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
-----	------	----------	--------	----------	------	---------	--------	-------	-----------------------------

VH2D-03-01	No	2026-05-28	18			101325	293	0	-
------------	----	------------	----	--	--	--------	-----	---	---

Group: VH2D-04

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
-----	------	----------	--------	----------	------	---------	--------	-------	-----------------------------

VH2D-04-01	No	2026-05-28	18			101325	293	0	-
------------	----	------------	----	--	--	--------	-----	---	---

Group: VH2D-05

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
-----	------	----------	--------	----------	------	---------	--------	-------	-----------------------------

VH2D-05-01	No	2026-05-29				101325	293	0	-
------------	----	------------	--	--	--	--------	-----	---	---

Group: VH2D-06

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
-----	------	----------	--------	----------	------	---------	--------	-------	-----------------------------

VH2D-06-01	No	2026-05-29				101325	293	0	-
------------	----	------------	--	--	--	--------	-----	---	---

EXDA DAQ Systems Export

Project: VH2D-Project

Responsible Researcher: PhD Student Javier I. Camacho

Total DAQ Systems: 5

Reference PDF: LU-DBI Measurement Chain (frontend/public/LU-DBI_MeasurementChain_v001.pdf)

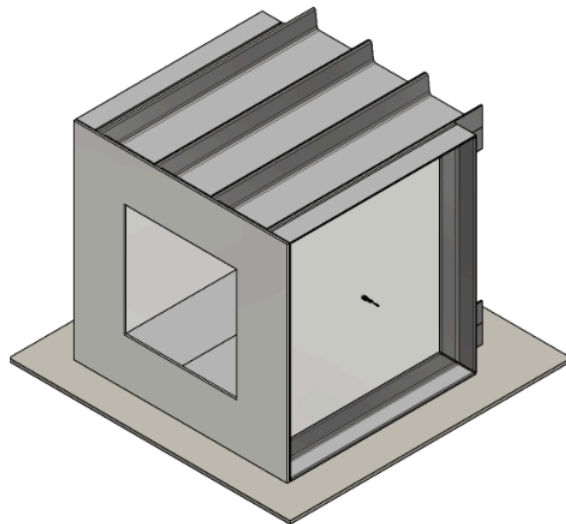
Reference PDF: Mixture Sampling Sub-System (frontend/public/MixtureSampling_Sub-System_000.pdf)

Generated: 2026-05-24 18:08:55

Name	Quantity	Vendor	Model	Serial	Fs (Hz)	Ch	Owner	Last Cal.	Cal Cert ID	Active	Notes
DAQ-1	pressure	Kistler	-	-	-	8	LU	-	-	Yes	-
DAQ-2	pressure	Kistler	-	-	-	4	DBI	-	-	Yes	-
DAQ-3	pressure	Kistler	-	-	-	4	Kistler	-	-	Yes	-
DAQ-4	pressure	Kistler	-	-	-	8	Kistler	-	-	Yes	-
H2CM	concentration	JCT	JTCD-300	14362	1	-	LU	2025-07-21	-	Yes	-

DAQ Reference Diagram - LU-DBI Measurement Chain

Source: frontend/public/LU-DBI_MeasurementChain_v001.pdf

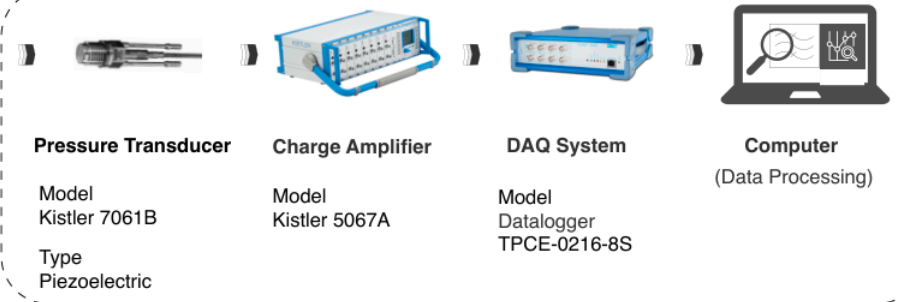


Explosion Chamber

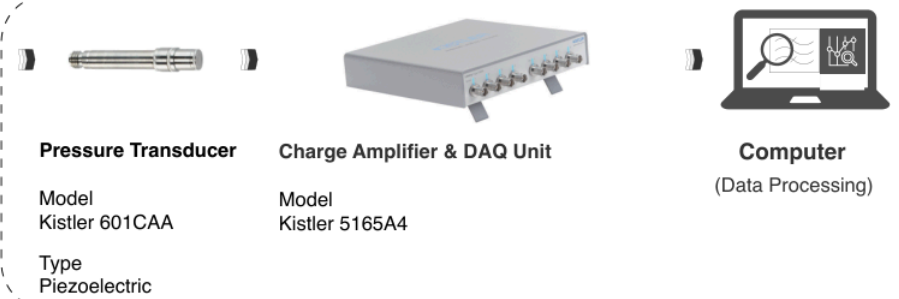
Dimensions (Internal): 900 x 900 x 900 [mm]

Material: Steel

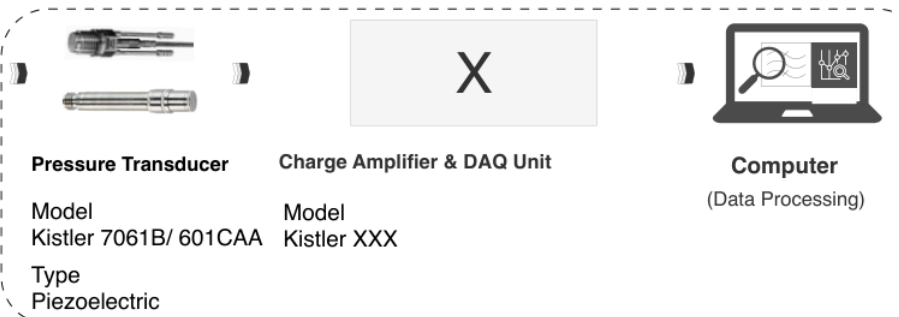
LU-Measurement Chain



DBI-Measurement Chain

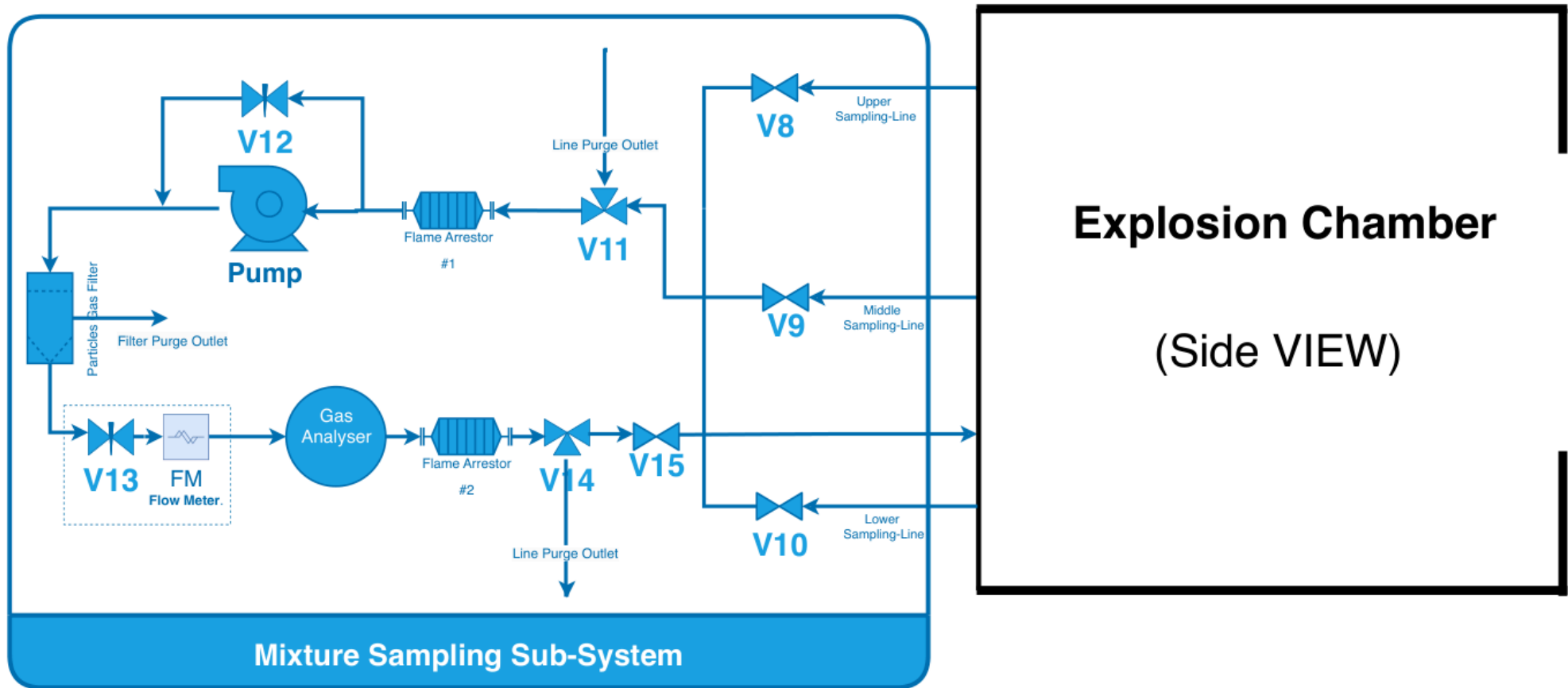


Kistler-Measurement Chain



DAQ Reference Diagram - Mixture Sampling Sub-System

Source: frontend/public/MixtureSampling_Sub-System_000.pdf



EXDA Sensors Mapping Export

Project: VH2D-Project

Responsible Researcher: PhD Student Javier I. Camacho

Total Groups: 7

Total Sensor IDs: 11

Total Sensor Mappings: 12

Sensor Mounting Reference File: frontend/public/SensorMountingLocation-000.pdf

Generated: 2026-05-24 18:08:55

Group: Prep

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivity	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
Prep	P1	pressure	DAQ-1	CH1	m123	75 pC/bar	1L	(1,1,1)	recessed	Yes	No	-	Complete
Prep	P2	pressure	DAQ-1	CH2	-	- pC/bar	left wall	(-, -, -)	flush	Yes	No	-	Missing/Invalid (missing serial; invalid sensitivity)
Prep	P3	pressure	DAQ-1	CH3	-	- pC/bar	blind/control channel	(-, -, -)	other	Yes	Yes	-	Missing/Invalid (missing serial; invalid sensitivity)
Prep	P4	pressure	DAQ-1	CH4	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	Missing/Invalid (missing serial; invalid sensitivity)
Prep	P5	pressure	DAQ-2	CH1	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	Missing/Invalid (missing serial; invalid sensitivity)
Prep	P6	pressure	DAQ-2	CH2	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	Missing/Invalid

Prep	P7	pressure	DAQ-2	CH3	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	valid (missing serial; invalid se nsitivity; Missing/In
Prep	P8	pressure	DAQ-3	CH1	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	valid (missing serial; invalid se nsitivity; Missing/In
Prep	Trigger-1	voltage	DAQ-1	CH5	-	- pC/bar	-	(-, -, -)	other	Yes	No	M-Duino	Complete
Prep	Trigger-2	voltage	DAQ-2	CH4	-	- pC/bar	-	(-, -, -)	other	Yes	No	M-Duino	Complete
Prep	Trigger-3	voltage	DAQ-3	CH2	-	- pC/bar	-	(-, -, -)	other	Yes	No	M-Duino	Complete

Group: VH2D-01

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivity	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-01	P8	pressure	Kistler LabAmp	-	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	Missing/In valid (missing DAQ channel; missing serial)

Group Reference Note: same configuration as preparation group

Group: VH2D-02

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivity	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-02	-	-	-	-	-	-	-	-	-	-	-	-	Reference only (no

Page 9/12

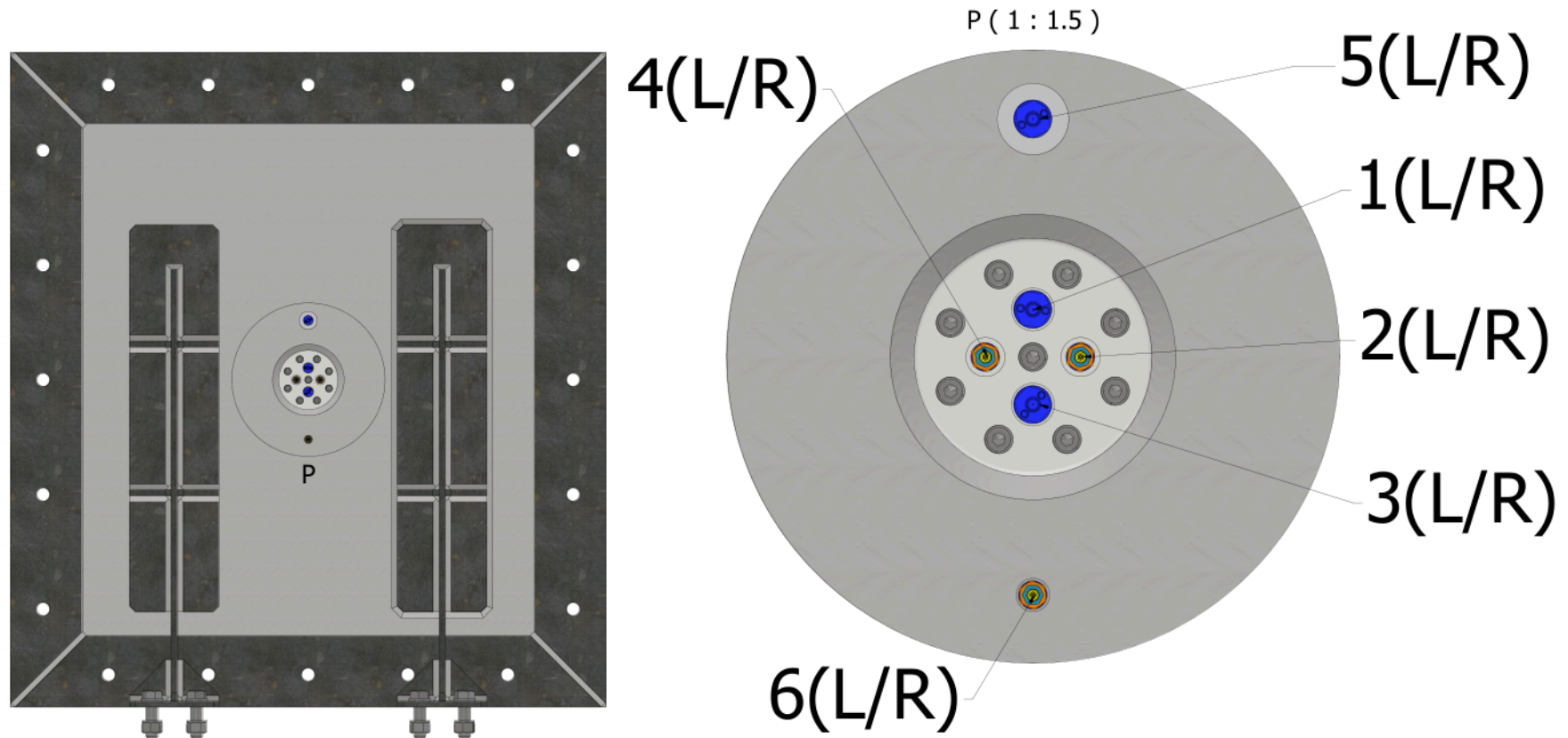


Group	Reference	Note	same configuration as preparation group	sensor changes)

Sensor Mounting Reference Diagram

Source: frontend/public/SensorMountingLocation-000.pdf

Pressure Sensor Location Mapping Overview



EXDA Gas Mixing Export

Project: VH2D-Project

Responsible Researcher: PhD Student Javier I. Camacho

Total Records: 0

Generated: 2026-05-24 18:08:56

MATLAB Verified: Yes

Verification File A: scripts/GasMixingVerificationFiles/H2_MFC_Fill_Calculator_v000.m

Verification File B: scripts/GasMixingVerificationFiles/AuxFcn_H2_MFC_FillCalculator_000.m

Shared Chamber Setup (applies to all runs unless noted otherwise):

L=- m, W=- m, H=- m, Pipes+=- m³, Hotwire+=- m³, Welded=- m³, Bolts=- m³, Vcorr=- L

Core Gas Mixing Results

Group	Run/Test	H2%	Pch(Pa)	Tch(K)	H2(g)	H2inj(L)	MFC	t(s)	t(min)	Updated
-------	----------	-----	---------	--------	-------	----------	-----	------	--------	---------

-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----